PROBLEM 1.7

Free Fall Motion under Gravity

Original statement · ProblemsCh1.pdf, p. 2

Newton II • Successive Integration • Kinematics

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

> **Problem statement:** Consider an object in free fall from an initial point $y(0) = y_0$ and with initial velocity 0, under the action of gravity and neglecting air friction. Write a differential equation for the position of the object $y(t)$ and solve it.

PHASE 2: FUNDAMENTAL FORMULATION

Coordinate y oriented vertically upward. Forces: $\vec{F}_g = -mg\hat{j}$. Newton's second law: $m\frac{d^2y}{dt^2} = -mg \implies \frac{d^2y}{dt^2} = -g$. Initial data: $y(0) = y_0$, $y'(0) = 0$.

> METHOD OF THE CLASS NOTES [NOTES, P. 2]
> The notes write the law as $-mg = m\,y''(t)$, a second-order linear ODE, and integrate twice: $y = -\iint g\,dt\,dt$. With $y_0 = 0$ the result below is the $-\frac{1}{2}gt^2$ of the notes.

PHASE 3: STEP-BY-STEP DERIVATION

Stage 1: Velocity Integration:

$$\frac{d}{dt}\left(\frac{dy}{dt}\right) = -g \implies \int_0^t \frac{d}{ds}\left(\frac{dy}{ds}\right) ds = \int_0^t -g\,ds$$

$$\frac{dy}{dt}(t) - \frac{dy}{dt}(0) = -gt \implies v(t) = \frac{dy}{dt} = -gt \qquad (1.7.1)$$

Stage 2: Position Integration:

$$\int_0^t \frac{dy}{ds}\,ds = \int_0^t -gs\,ds \implies y(t) - y(0) = \left[-\frac{1}{2}gs^2\right]_0^t$$

$$y(t) - y_0 = -\frac{1}{2}gt^2 \implies \mathbf{y(t) = y_0 - \frac{1}{2}gt^2} \tag{1.7.2}$$

PHASE 4: FINAL RESULT & ENERGY CONSERVATION

BOXED RESULT

$$y(t) = y_0 - \frac{1}{2}gt^2, \quad v(t) = -gt$$

Energy Check: $E = \frac{1}{2}mv^2 + mgy = \frac{1}{2}m(-gt)^2 + mg(y_0 - \frac{1}{2}gt^2) = mgy_0 = \text{constant}$. Mechanical energy is strictly conserved.